NEXUS UNIVERSAL COLLECTIBLES SYSTEM

Complete System Model - January 2026

Desktop → Scanner → Marketplace | B2B + B2C | Hardware + SaaS

EXECUTIVE SUMMARY

NEXUS is a patent-pending universal collectibles platform combining:

- **Hardware:** Enclosed scanning chamber with robotic arm + AI cameras
- **Software:** Desktop application with 5-module dashboard
- **Marketplace:** B2B/B2C trading platform with 5% transaction fees
- **AI:** Pre-grading prediction, OCR recognition, cross-category learning

The Innovation: One hardware investment serves the entire $248B collectibles market through a "zero-sort" library system that eliminates physical sorting infrastructure.

Metric	Value
Patent Status	Provisional filed Nov 27, 2025 (71 claims)
Current Customers	3 shops waiting (Duck Hunter signed $5K)
Hardware BOM	~$800
SaaS Price	$149-450/month
Target Market	30,000+ US collectible shops
5-Year Revenue	$469M (blended projection)

SYSTEM ARCHITECTURE

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NEXUS ECOSYSTEM
├── SCANNER HARDWARE
│   ├── Enclosed Chamber (PC case, matte black)
│   ├── Camera System (CZUR + OwlSight 64MP × 2)
│   ├── LED Lighting (WS2812B, 3000K-6500K)
│   ├── Robotic Arm (NEMA 17 + 5:1 gearbox)
│   └── Pi 5 Cluster (Brok, Snarf, Zultan)
│
├── DESKTOP SOFTWARE (NEXUS V2)
│   ├── Collection Management
│   ├── Hardware Integration
│   ├── AI Analytics
│   ├── Marketplace Portal
│   └── Business Intelligence
│
├── MARKETPLACE PLATFORM
│   ├── B2B Storefronts
│   ├── B2C Browse/Buy
│   └── 5% Transaction Fees
│
└── AI LAYER
    ├── 5-Point ID Recognition
    ├── Visual Fingerprinting
    ├── PSA Grade Prediction (93% accuracy)
    └── Cross-Category Learning
```

HARDWARE COMPONENTS

Professional Scanner Station (B2B)

Component	Specs	Cost
Raspberry Pi 5 × 3	Edge computing cluster	$240
OwlSight 64MP × 2	Stereo vision cameras	$120
CZUR Scanner	Front/overhead capture	$180
NEMA 17 + Gearbox	Base rotation (5:1 planetary)	$45
ZK-SMC02	Motor controller w/ display	$28
WS2812B LEDs	Ring + strip lighting	$30
Enclosure	PC case or custom	$100
Cables, PSU, misc	Power and data	$50
TOTAL BOM		**~$800**

Consumer Handheld (B2C - Year 3)

Spec	Value
Size	5" × 3" × 1"
Weight	< 8 oz
Camera	12MP+ macro
Battery	8+ hours
Price	$249 retail

ZERO-SORT LIBRARY SYSTEM

The Innovation

Traditional: Sort by category → set → name → number (bins, dividers, infrastructure)

NEXUS: Scan → Assign call number → Place anywhere → Digital lookup

Call Number Format

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AA-0001, AA-0002, ... AA-9999
AB-0001, AB-0002, ... AB-9999
...
ZZ-9999

Total Capacity: 6,759,324 items
```

Benefits

Factor	Traditional	NEXUS Zero-Sort
Setup time	Hours	Zero
Infrastructure	Bins, dividers	Any storage
Finding items	Search bins	Direct lookup
Scaling	Complexity grows	Linear
Error rate	High (missorts)	Near zero

Current library: 26,850+ cards cataloged using Scryfall IDs

THREE SCANNING METHODS

Method	Speed	Accuracy	Use Case
Single Type Bulk	100+/hr	85-90%	Known category, high volume
Mixed Bulk	80-90/hr	85-90%	Estates, unknown collections
Pre-Grading	50-60/hr	95-98%	High value, grading submission

Method 1: Single Type Bulk

1. Select card type (MTG/Pokemon/etc.)
2. CZUR captures front
3. Tesseract OCR → 5-Point ID
4. Assign call number → Save → NEXT

Method 2: Mixed Bulk

1. OwlSight captures BACK → Detect type
2. CZUR captures front
3. Tesseract OCR → 5-Point ID
4. Assign call number → Save → NEXT

Method 3: Pre-Grading

1. Both OwlSight cameras (64MP front + back)
2. EasyOCR full analysis
3. PSA grading algorithm (centering, corners, edges, surface)
4. Grade prediction (1-10 scale)
5. Condition report → Save with grade

OCR & RECOGNITION

5-Point ID System

Region	Purpose
1. Card Name	Primary identifier
2. Set Symbol	Visual confirmation
3. Mana Cost	Card validation
4. Type Line	Card type
5. Collector Info	Unique ID (SET 123/456)

Dual OCR Engine

Engine	Speed	Accuracy	Use Case
Tesseract	0.2s	85%	Bulk scanning
EasyOCR	1.5s	95%	Pre-grading

Visual Fingerprinting

- pHash + aHash + dHash combined
- Hamming distance < 10 = match
- Enables counterfeit detection

AI PRE-GRADING SYSTEM

PSA Grade Prediction

Component	Weight	Analysis
Centering	40%	Border measurement
Corners	25%	Sharpness/wear
Edges	20%	Straightness/chipping
Surface	15%	Scratches/defects

Grade Scale

Grade	Label	Score
10	Gem Mint	99.5%+
9	Mint	95%+
8	NM-MT	90%+
7	Near Mint	85%+

Accuracy: 93% within 1 PSA grade

ROI Calculator

```
Card: Black Lotus (Alpha)
Estimated: PSA 8
Raw Value: $25,000
Graded Value: $85,000
Grading Fee: $300
Profit: $59,700 → SUBMIT ✓
```

BUSINESS MODELS

B2B: Professional Scanner

Tier	Monthly	Categories
Base	$149	1 category
Multi	$249	3 categories
Universal	$449	All categories

Hardware included (lease) or purchase $2,500 + $99/mo

B2C: Consumer Handheld (Year 3)

Model	Price	Features
Lite	$149	Basic, 1 category
Home	$249	Full, all categories
Pro	$399	Pre-grading, marketplace

Unit Economics

Metric	Value
BOM	$800
Avg price	$250/mo
Break-even	3.2 months
Gross margin	79-85%
LTV	$6,500
LTV:CAC	6.5:1

PRIOR ART & COMPETITORS

Current Solutions (All Lacking)

Solution	Limitation
Manual inspection	Slow, inconsistent, expensive
Phone apps	Uncontrolled lighting, 60-70% accuracy
PSA Genamint	No hardware, 85% accuracy, B2C only
DIY setups	No support, 60-70% accuracy

Direct Competitors: NONE

No competitor offers hardware-based pre-grading for collectibles. NEXUS is creating a new category.

Indirect Competitors

Competitor	Type	Threat
PSA Genamint	Mobile app	Medium
Human experts	Manual	Low
TCGPlayer	Marketplace only	Low
Future PSA scanner	Not built yet	Medium-High

DIFFERENTIATION SUMMARY

Factor	TCGPlayer	PSA App	Human	NEXUS
Hardware	❌	❌	❌	✅
Categories	TCG only	Cards	Any	✅ Universal
Speed	N/A	30/hr	10-15/hr	✅ 100+/hr
Accuracy	N/A	85%	80-90%	✅ 93%
Pre-grading	❌	Basic	Subjective	✅ AI
Marketplace	10.25%	❌	❌	✅ 5%
Zero-Sort	❌	❌	❌	✅ Patent
Cross-AI	❌	❌	❌	✅ Patent
Patent	❌	❌	❌	✅ 71 claims

FINANCIAL PROJECTIONS

5-Year Summary (Blended)

Year	B2B	B2C	Revenue	Profit
2026	275	2,500	$4.2M	-$150K
2027	1,485	22,500	$24.2M	$3.3M
2028	4,485	91,500	$80.6M	$17.7M
2029	9,550	271,500	$213.5M	$64M
2030	17,000	627,500	$469M	$141M

Revenue Streams (Year 5)

Stream	Amount
B2B Hardware/Subscriptions	$67M
B2C Hardware/Subscriptions	$179M
Marketplace Fees	$38M
AI Grading	$22M
API/Enterprise	$8M
Total	**$469M**

PATENT CLAIMS (71 TOTAL)

Category	Independent	Dependent	Total
System	15	15	30
Method	8	12	20
Apparatus	4	11	15
AI/ML	2	4	6
TOTAL	**29**	**42**	**71**

Key Claims

- Claim 1: Enclosed scanning chamber
- Claim 2: Articulated robotic arm
- Claim 3: Multi-region extraction (5-Point ID)
- Claim 5: Cross-category AI learning
- Claim 10: AI grading prediction
- Claim 31: Universal scanning method
- Claim 51: Chamber apparatus
- Claim 66: Visual fingerprint system

Filing Status

- Provisional: Nov 27, 2025 ✓
- Non-provisional deadline: Nov 27, 2026
- Attorney: Lerner David (Westfield, NJ)
- USPTO fees (micro): ~$2,432

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Patent: USPTO Provisional #63/926,477

Next milestone: Duck Hunter deployment Friday